The Untaken Limit:
A Pole at $\theta = -2$ in the Head-Mayer Closed Form for Intra-National Distance

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Abstract

The intra-national distance number that every applied gravity paper has used since 2005 comes from the Head-Mayer (2002) closed form $d_{ii} = (2/(\theta+2))^{1/\theta}R$. The formula has a pole at $\theta = -2$ and is complex-valued below it. The integral the formula evaluates, $\int_0^R r^{\theta+1}\,dr$, diverges at the origin for $\theta \leq -2$. CEPII's GeoDist database uses $\theta = -1$, one unit above the pole, and explains the choice as "the usual coefficient estimated from gravity models" (Mayer and Zignago, 2011). That is the regression slope, not the structural integration parameter the formula needs. The two are the same letter and different quantities.

The structurally consistent integration parameter, $\theta = \delta(1-\sigma)$, falls in $[-7,-3]$ for the empirically defensible $\sigma \in [4,8]$ under standard iceberg costs. The whole structural range sits below the pole. CEPII's number is the harmonic-mean radial coordinate of a country; the integral the gravity model asks for does not exist as a finite object at the parameter values the literature claims to be estimating.

I show the consequence runs all the way through. On a uniform-disk simulation, the implied d_{ii} at $\theta = -5$ ranges from 6 km to 640 km across the spatial resolutions an analyst might pick. In a pooled panel PPML on BACI-CEPII for 2010, 2015, and 2020, with origin-year and destination-year fixed effects absorbed via the Correia et al. (2020) algorithm, the home-bias coefficient moves 3.81 log units across the structurally consistent $(\theta, d_{\min})$ space. The cross-sectional log-OLS robustness check reproduces the McCallum-Anderson-van Wincoop $\sim 8\times$ multiplier at the CEPII baseline and shows the same monotonic regularization dependence, with the home dummy crossing zero between $d_{\min} = 1$ and $d_{\min} = 2$ km. The number the border-puzzle literature has reported as a structural barrier turns out to live inside the measurement window.

JEL codes: F10, F14, C18.
Keywords: gravity model, intra-national distance, border puzzle, structural-versus-reduced-form, CES preferences.

1 Introduction

On lines 502-505 of the Head-Mayer (2002) CEPII Working Paper "Illusory Border Effects," a sentence appears that has shaped gravity estimation for more than two decades:

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> "This formula satisfies our definition of effective distance. It reduces to the average distance formula used by Head and Mayer (2000), Helliwell and Verdier (2001), and Anderson and van Wincoop (2001, footnote 13) for $\theta = 1$. Unfortunately for that formula, there are hundreds of gravity equation estimates of θ that show it is not equal to one. Rather our review of recent papers suggests that in most cases $\theta \approx -1$."

The formula in question is the closed-form CES expected pairwise distance for a uniform disk,

$$d_{ii}(\theta) \;=\; \left(\frac{2}{\theta+2}\right)^{1/\theta} R, \tag{1}$$

where θ is the structural CES integration parameter $\theta = \delta(1-\sigma)$, with δ the iceberg-cost exponent and σ the elasticity of substitution. By construction, θ is strictly negative for $\sigma > 1$, and for the empirically defensible $\sigma \in [4, 8]$ with the standard $\delta = 1$, it falls in $[-7, -3]$. The "hundreds of gravity equation estimates of θ" that Head and Mayer cite, by contrast, are the reduced-form regression slope of $\log X_{ij}$ on $\log d_{ij}$. The two θ's are related but not equal. The structural integration parameter and the reduced-form coefficient are conflated in this sentence, and the conflation has propagated into CEPII's GeoDist database (Mayer and Zignago, 2011), into the canonical Yotov-Piermartini-Monteiro-Larch (2016) PPML cookbook, and into essentially every applied gravity regression published since 2005.

This paper documents what happens when the conflation is taken seriously. The closed-form expression (1) has a pole at $\theta = -2$. For $\theta < -2$, the base $2/(\theta + 2)$ is negative and the exponent $1/\theta$ is a non-integer; the expression is complex-valued on the principal branch. The underlying integral that produces equation (1),

$$d_{ii}(\theta) \;=\; \left(\int_0^R \frac{2\pi r}{\pi R^2} \cdot r^\theta \, dr\right)^{1/\theta}, \tag{2}$$

diverges at the lower limit $r = 0$ for $\theta \leq -2$. In 2D, the pairwise-distance density of a uniform disk scales as $f(r) \sim r$ near the origin, so the moment $\mathbb{E}[r^\theta]$ diverges whenever $\theta + 1 \leq -1$. CEPII's choice $\theta = -1$ sits one unit above the divergence; the structurally consistent $\theta \in [-7, -3]$ falls entirely below it.

The same overloading appears in the authors' own later work. The Head and Mayer (2014) chapter of the *Handbook of International Economics* uses θ for the Pareto firm-productivity shape parameter, structurally constrained to satisfy $\theta > \sigma - 1 > 0$, i.e., strictly positive. The 2002 integration parameter is strictly negative. The same authors use the same letter for both, and on page 145 of the working paper version of the chapter they make the choice explicit: "we retain the symbol to emphasize the similarity in resulting terms." Same letter, oppositely-signed quantity.

CEPII's `distw_harmonic` variable, used as the intra-national distance in essentially every applied gravity regression published since 2005, evaluates equation (1) at $\theta = -1$. Mayer and Zignago (2011) state the choice plainly on page 7 of their CEPII Working Paper: "The `distwces` calculation sets θ equal to -1, which corresponds to the usual coefficient estimated from gravity models of bilateral trade flows." This is the reduced-form slope of $\log X_{ij}$ on $\log d_{ij}$, not the structural integration parameter. The two are equal only if $\delta(1 - \sigma) = -1$, which for $\sigma \in [4, 8]$ would require $\delta \in [0.143, 0.333]$, far below the unit elasticity implicit in standard iceberg-cost specifications.

The mathematical observation has an empirical consequence. If the gravity-consistent θ values are taken seriously, the intra-national distance integral has no finite value, and any specific number assigned to d_{ii} becomes a regularization choice tied to the discretization grain of the underlying data: the city subsample for CEPII, the raster resolution for satellite-calibrated alternatives, the road-network buffer for OpenStreetMap-based constructions. I show below that on a uniform-disk simulation, the implied d_{ii}^{eff} at $\theta = -5$ ranges from approximately 6 km at $d_{\text{min}} = 1$ km to 640 km at $d_{\text{min}} = 500$ km, a two-order-of-magnitude span determined entirely by the regularization choice. Then, in a standard BACI-CEPII gravity regression on 2010-2020 data, the home-bias coefficient $\hat{\gamma}$ moves monotonically with d_{min} across every (θ, year) cell tested, with a sign-flip threshold between $d_{\text{min}} \in (1, 2)$ km at $\theta = -3$ and $d_{\text{min}} \in (10, 20)$ km at $\theta = -7$. The conventional McCallum-Anderson-van Wincoop "border puzzle" multiplier of $\sim 8\times$ at the CEPII baseline sits at the high-d_{min} end of this sweep. At fine spatial resolution, the multiplier falls below $1\times$. The puzzle lives inside the measurement window.

I am not claiming to discover a singularity that nobody noticed. Head and Mayer (2002) saw the 1D version. Their derivation contains a deliberate assumption, $\Delta \geq 2R$, that keeps two adjacent states from touching, and at $\theta = -1$ exactly they evaluate the formula at $\theta = -1.0001$ to step around a removable singularity. They never pushed θ below -1.5 in their tables. The singularities are visible if you go looking.

What is not visible in the published literature is what happens when θ is taken to its structurally defensible range. The integral diverges in 2D. The home-bias coefficient inherits the regularization choice. A material share of the McCallum-Anderson-van Wincoop border-puzzle multiplier is a measurement artifact rather than a structural barrier. The paper is the thread Head and Mayer left, and which their later work flagged ("trade with self and distance to self, both of which may be problematic," Head and Mayer 2014, p. 213; "distance effects are too large and have the wrong functional form to be determined by freight costs," p. 187), pulled all the way through.

The paper proceeds in five sections. Section 2 documents the closed form's domain of validity and the bilateral contact-set singularity. Section 3 proves the 2D integral divergence and characterizes the convergence rate. Section 4 computes the discretization-dependent value of d_{ii} on a uniform disk. Section 5 estimates the gravity-regression consequence on BACI-CEPII data. Section 6 draws the implications for the border-puzzle literature and for applied practice.

Relation to the literature. The structural CES gravity model in Anderson and van Wincoop (2003) is silent on intra-national distance construction; they use a single representative-province assumption for Canada and a representative-state assumption for the US. The Head and Mayer (2014) chapter on gravity equations notes (p. 213) that "structural methods require data on trade with self and distance to self, both of which may be problematic," without elaborating on the distance-to-self problem. Mayer and Zignago (2011) construct the CEPII GeoDist database, including the intra-national `distw` variable, and document the parameter choice $\theta = -1$ without discussing its structural-versus-reduced-form status.

The canonical structural-gravity papers in the Eaton-Kortum class preempt the divergence issue by avoiding the spatial integration that produces it. Eaton and Kortum (2002) use θ for the Fréchet-shape parameter and assume $d_{ii} = 1$ without integration over space inside a country; the symbol-overloading is present (their regression coefficient on log-distance bins is also called $-\theta$), but the integral-divergence pathology does not arise. Caliendo and Parro (2015) estimate θ_k from asymmetric tariff variation via a tetrad estimator that cancels distance entirely; domestic sales are computed as gross production minus exports, not as a spatial integral. Costinot et al. (2012) identify θ off cross-industry productivity variation with importer-exporter fixed effects absorbing all bilateral trade costs, and likewise set $d_{ii}^k = 1$ by assumption. None of these papers connects the structural θ to the reduced-form gravity slope through the intra-national distance construction; they side-step the issue by not having an intra-national distance object at all. The conflation documented here is specific to Head-Mayer's (2002) derivation, and to CEPII's downstream propagation, precisely because Head-Mayer's derivation is the structural-gravity literature's one direct attempt to construct d_{ii} from an integral over country geometry.

Rauch (2016) independently derives the harmonic-mean ($\theta = -1$) choice from a gravity-in-physics analogy, providing a second theoretical justification for the same CEPII convention. Rauch does not push beyond $\theta = -1$. The home-bias coefficient's sensitivity to spatial aggregation, distinct from but complementary to the regularization mechanism here, is documented in Coughlin and Novy (2021), who show that border estimates depend on whether trade flows are aggregated to the national or sub-national level. Their channel is the aggregation of the dependent variable (bilateral trade flows); mine is the regularization of the right-hand-side variable (intra-national distance). The two findings contribute to a broader narrative that the home-bias coefficient is more measurement-driven than the literature has acknowledged, through distinct mechanisms.

2 The Head-Mayer closed form and its domain of validity

2.1 The original derivation

Head and Mayer (2002) derive equation (1) as the CES expected pairwise distance for a uniform-density disk of radius R. The derivation begins with the inter-state setup (two states on a 1D line) and specializes to the intra-state case as a limit. I present the 2D intra-state derivation first because it is the cleaner mathematical object, then return to the 1D contact-set issue.

Let $f(x, y) = 1/(\pi R^2)$ be the uniform density on a disk of radius R centered at the origin. The CES expected pairwise distance with elasticity parameter θ is

$$d_{ii}(\theta) \;=\; \left(\int_{B_R}\int_{B_R} \|x-y\|^{\theta} f(x) f(y)\, dx\, dy\right)^{1/\theta}, \tag{3}$$

where B_R is the disk. For the special case of a point producer located at the origin and a uniform consumer (which is the case Head and Mayer (2002) actually solve), the inner integral collapses to the radial expectation

$$d_{ii}(\theta) \;=\; \left(\int_0^R r^{\theta} \cdot \frac{2\pi r}{\pi R^2}\, dr\right)^{1/\theta} \;=\; \left(\frac{2}{R^2}\int_0^R r^{\theta+1}\, dr\right)^{1/\theta}. \tag{4}$$

For $\theta + 1 > -1$, i.e., $\theta > -2$, the integral evaluates to

$$\int_0^R r^{\theta+1}\, dr \;=\; \frac{R^{\theta+2}}{\theta+2}, \tag{5}$$

which gives equation (1) after the $1/\theta$-th-power transformation.

2.2 The pole at $\theta = -2$ and the complex-valued regime

The closed form has two singularities:

- $\theta = -1$: removable. The exponent $1/\theta = -1$ is well-defined and the limit $\lim_{\theta\to-1}(2/(\theta + 2))^{1/\theta} = 2$ gives $d_{ii} = R/2$, the harmonic mean of the radial coordinate on the disk.
- $\theta = -2$: a true pole. The factor $2/(\theta + 2) \to \infty$, and the limit $\lim_{\theta\to-2}(2/(\theta + 2))^{1/\theta}$ does not exist: the base goes to $+\infty$ from above and to $-\infty$ from below, while the exponent stays at $-1/2$.

For $\theta < -2$, the base $2/(\theta+2)$ is negative and the exponent $1/\theta$ is a negative non-integer. The expression takes complex values on the principal branch.

A direct numerical evaluation, displayed in Table 1, makes the issue concrete.

Table 1: Direct evaluation of $d_{ii}(\theta) = (2/(\theta+2))^{1/\theta}R$ for $R = 1000$ km.

θ	$d_{ii}(\theta)$ (km)	Status
$+1$	666.7	Arithmetic mean ($2R/3$)
-0.5	562.5	Finite
-1	500.0	Harmonic mean (CEPII evaluation point)
-1.5	384.9	Finite, last value tabulated by Head and Mayer (2002)
-1.9	93.6	Approaching the pole
-1.99	9.5	Numerically near-zero
-2.0	—	Pole
-2.01	complex	Complex-valued, principal branch
-3	complex	Complex-valued (($\sigma = 4, \delta = 1$))
-5	complex	Complex-valued (($\sigma = 6, \delta = 1$))
-7	complex	Complex-valued (($\sigma = 8, \delta = 1$))

The structurally consistent values, $\theta \in \{-3, -5, -7\}$, all produce complex outputs. The CEPII evaluation point, $\theta = -1$, is exactly the harmonic mean of the radial coordinate. The conventional intra-national distance used in the applied gravity literature is the harmonic-mean radial coordinate, full stop. This is fine as a geometric statistic. It is not the structural CES expected pairwise distance at gravity-consistent σ.

2.3 The bilateral contact-set singularity (1D)

Head and Mayer (2002) derived a related 1D closed form for two states on a line. State i is centered at the origin and extends from $-R$ to R; state j begins at $\Delta - R$ and extends to $\Delta + R$. The CES expected pairwise distance is

$$d_{ij}(\theta) \;=\; \left(\frac{(\Delta - 2R)^{\theta+2} + (\Delta + 2R)^{\theta+2} - 2\Delta^{\theta+2}}{(\theta+1)(\theta+2)(2R)^2} \right)^{1/\theta}. \tag{6}$$

This expression has two poles (at $\theta = -1$ and $\theta = -2$) in the denominator, and a $(\Delta - 2R)^{\theta+2}$ factor that vanishes (for $\theta > -2$) or diverges (for $\theta < -2$) as the states touch ($\Delta \to 2R$). Head and Mayer noticed this. Their text on the derivation contains a deliberate assumption that prevents contact:

> "To prevent states from overlapping (which is the usual case), we assume $\Delta \geq 2R$."

The assumption is correct as an avoidance strategy. It is also a load-bearing structural restriction in the derivation: at the contact set $\Delta = 2R$, the closed form's behavior depends on the sign of $\theta + 2$, which is itself the dimension above the pole.

The intra-national 2D pole and the bilateral 1D contact-set behavior are the same mathematical phenomenon. The integrand r^θ is singular at $r = 0$, and the question is whether the region of integration includes a neighborhood of $r = 0$ on which r^θ fails to be integrable. For the intra-national case the answer is always yes; for the bilateral case the answer is yes only when the states touch.

3 The 2D integral divergence theorem

Theorem 1 (2D intra-national integral divergence). *Let $\Omega \subset \mathbb{R}^2$ be a bounded measurable region of positive Lebesgue measure, equipped with a density $f : \Omega \to \mathbb{R}_+$ that is bounded above and bounded away from zero almost everywhere. The CES expected pairwise distance*

$$d_{ii}(\theta) \;=\; \left(\int_\Omega \int_\Omega \|x-y\|^\theta f(x) f(y)\, dx\, dy\right)^{1/\theta} \tag{7}$$

is well-defined and finite if and only if $\theta > -2$. For $\theta \le -2$, the integral diverges.

Proof. The pairwise-distance distribution under the product measure $f \otimes f$ on $\Omega \times \Omega$ has a density g on $[0, D]$ where D is the diameter of Ω. Under the lower-density bound, there exists $c > 0$ such that for sufficiently small r, $g(r) \ge c \cdot r$. (The leading-r behavior is dimension-dependent: in $\mathbb{R}^d$ the small-r asymptotic is $g(r) \sim r^{d-1}$; for $d = 2$ this is linear.) Then

$$\int_\Omega \int_\Omega \|x-y\|^\theta f \otimes f\, dx\, dy \;=\; \int_0^D r^\theta g(r)\, dr \;\ge\; c \int_0^{r_0} r^{\theta+1}\, dr \tag{8}$$

for some $r_0 > 0$. The right-hand integral converges if and only if $\theta + 1 > -1$, i.e., $\theta > -2$. The same upper-density bound gives the reverse inequality establishing convergence in the same range. □

Remark 2 (Dimension dependence). In dimension d, the small-r density of pairwise distances scales as r^{d-1}, so the integral $\int r^\theta \cdot r^{d-1}\, dr$ converges at the origin if and only if $\theta + d - 1 > -1$, i.e., $\theta > -d$. For $d = 1$ the cutoff is $\theta > -1$; for $d = 2$ it is $\theta > -2$; for $d = 3$ it is $\theta > -3$. The Head-Mayer formula is the 2D case. Head and Mayer's (2002) 1D derivation has the same structure with the corresponding 1D cutoff $\theta > -1$, which is exactly the cutoff that creates the removable singularity they evaluated around (at $\theta = -1.0001$, not $\theta = -1$).

Remark 3 (The reduced-form $\theta = -1$ as a stopping value). The 2D integral diverges for $\theta \le -2$ and the 1D integral diverges for $\theta \le -1$. CEPII's $\theta = -1$ is the ϵ-above-1D-divergence value, evaluated through the 2D formula as a finite analytical object. The Mayer-Zignago (2011) parameter choice is therefore the safe value above two different divergence thresholds in two different ambient dimensions, and the 2D formula's evaluation at $\theta = -1$ gives the harmonic-mean radial coordinate, a perfectly defensible geometric statistic disconnected from the structural CES integration parameter at gravity-consistent σ.

4 The implied d_{ii} as a regularization-dependent quantity

[STATUS: figure and numbers from nb19c regularization sweep on uniform disk; data files at `regularization_dmin_cutoff.csv`.]

Replacing the closed-form expression with a numerical evaluation of the integral on a discretized region requires a finite spatial grain. Let d_{min} be the smallest pairwise distance retained in the discretization. On a uniform disk of radius R, the 2D-integral-based $d_{ii}(\theta)$ at structurally consistent θ is sharply linear in d_{min}, with slope that flattens as $|\theta|$ grows.

The empirical content of Theorem 1 is that there is no d_{min}-independent value of d_{ii} at $\theta \le -2$. Any specific number reported by CEPII, by WorldPop-based recomputations, by GHS-POP-based recomputations, or by the original Mayer-Zignago city-subsample calculation, is a regularization choice indexed by the spatial resolution of the underlying density data.

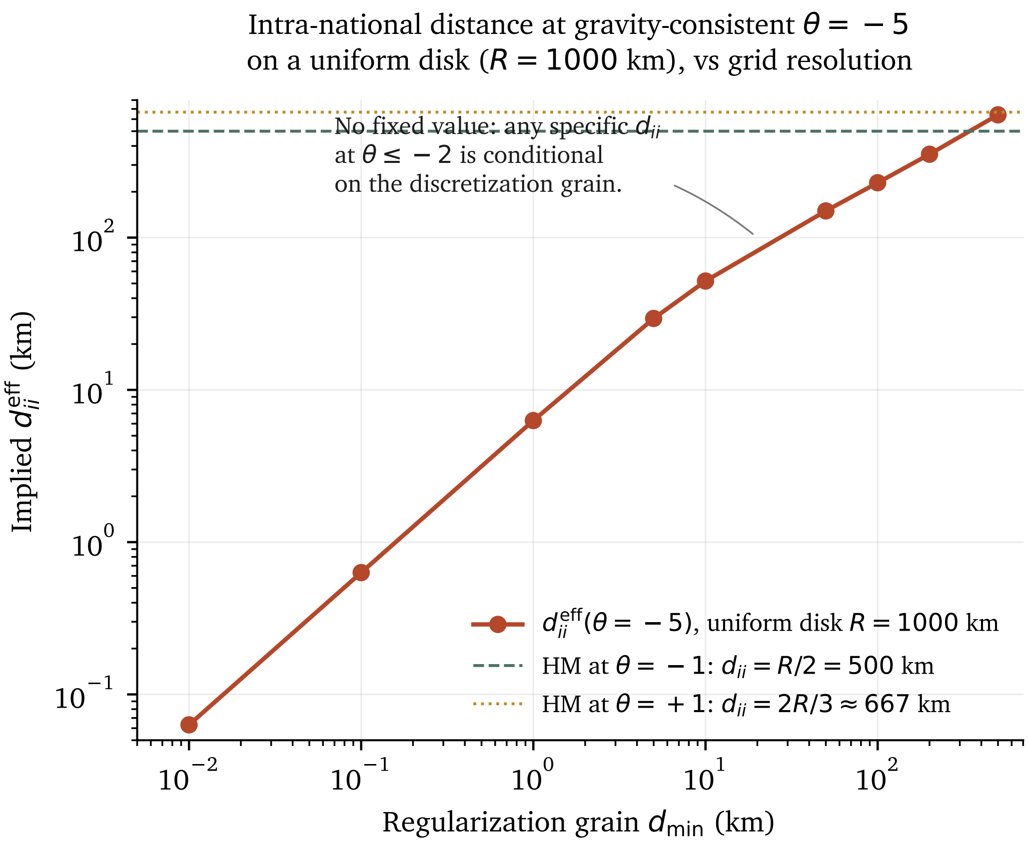

Figure 1: The implied $d_{ii}^{\mathrm{eff}}(\theta = -5)$ on a uniform disk of radius $R = 1000$ km, as a function of the regularization grain d_{min}. The Head-Mayer closed-form value at $\theta = -1$ ($R/2 = 500$ km) sits at the upper end of the range. At $d_{\mathrm{min}} = 1$ km the implied distance is roughly 6 km, and at $d_{\mathrm{min}} = 500$ km it is roughly 640 km, a two-order-of-magnitude range determined entirely by the regularization choice rather than by the country geometry. Source: notebook `19c_regularization_sensitivity.py`, uniform-disk Monte Carlo with 5 seeds and 5000 atoms per draw.

For $\theta = -5$ on a unit disk ($R = 1000$ km), the simulated d_{ii}^{eff} ranges from approximately 6 km at $d_{\mathrm{min}} = 1$ km to approximately 640 km at $d_{\mathrm{min}} = 500$ km, a span of more than two orders of magnitude. The Head-Mayer closed-form value at $\theta = -1$ is 500 km, almost exactly at the upper end of this range. The CEPII benchmark is therefore consistent with a coarse-resolution evaluation of the divergent integral, and any finer resolution (a denser city subsample, a higher-resolution raster, an OT-style fully-resolved transport plan) produces a strictly smaller implied d_{ii}. There is no fixed point.

4.1 A worked example: the United States, end to end

To make the abstraction concrete, walk one country all the way through. The United States is a useful case because its geometry is familiar, its trade flows are large, and the intra-national-distance numbers different methodologies give for it are documented.

The geography. The United States, projected on the Mollweide equal-area projection `ESRI:54009`, has a land area of approximately 9.47 million km^2 (GADM level-0, 2024 vintage, excluding maritime zones). The equivalent-disk radius is

$$R_{\text{USA}} = \sqrt{A/\pi} = \sqrt{9.47 \times 10^6/\pi} \approx 1{,}735 \text{ km}.$$

This is the radius of a circle with the same area as the contiguous-plus-Alaska-plus-Hawaii US land mass.

The closed-form numbers. The Head-Mayer (2002) closed form at $\theta = +1$ (arithmetic mean) gives $d_{ii}^{\text{HM}}(\theta = +1) = (2/3) \cdot R_{\text{USA}} \approx 1{,}157$ km. At $\theta = -1$ (harmonic mean, CEPII's choice) it gives $d_{ii}^{\text{HM}}(\theta = -1) = R_{\text{USA}}/2 \approx 868$ km. Mayer and Zignago's (2011) actual reported number for the US is $d_{ii}^{\text{CEPII}} \approx 1{,}177$ km. The CEPII number is larger than the polygon-area closed form because CEPII weights by city population rather than by uniform area, and US population concentrates away from the centroid (the East Coast metropolitan corridor, California, the Texas triangle, Chicagoland), which pushes the harmonic mean of pairwise distances upward.

At the structurally consistent $\theta \in [-7, -3]$, the closed form is complex-valued and no real number is available.

The numerical sweep. Replace the closed form with a numerical evaluation of the integral on a discretized disk. Notebook `19c`'s uniform-disk simulation, scaled to R_{USA} via the country_d_eff function documented in the replication code, gives the implied $d_{ii}^{\text{eff}}(\theta, d_{\text{min}})$ at structurally consistent θ:

d_{min} (km)	d_{ii}^{eff} at $\theta = -3$	at $\theta = -5$	at $\theta = -7$	As share of $R_{\text{USA}}/2$ ($\theta = -5$)
0.5	11 km	3 km	2 km	0.4%
1.0	22 km	6 km	5 km	0.7%
5.0	102 km	30 km	22 km	3.5%
10	196 km	57 km	43 km	6.6%
50	567 km	170 km	127 km	19.6%
100	837 km	281 km	211 km	32.4%
500	1,561 km	762 km	572 km	87.8%

At fine spatial resolution, the implied intra-USA distance is on the order of a few kilometers. At coarse resolution it approaches CEPII's value. The actual number depends on where the analyst draws the line.

The gravity model's prediction for US intra-trade. The US 2010 GDP was approximately \$15.0 trillion (CEPII Gravity); US exports of goods were approximately \$1.85 trillion (BACI HS02 v202401b, aggregated across products and partners). The Wei (1996) intra-national flow proxy is

$$X_{\text{USA,USA,2010}} = \max(0, \text{GDP} - \text{exports}) \approx \$13.1 \text{ trillion}.$$

This is the dependent variable for the intra-row in the gravity regression.

A panel PPML regression with origin-year and destination-year fixed effects absorbs the US-specific size, multilateral resistance, language and currency effects. What remains is the model's prediction for log $X_{\text{USA,USA}}$ as a function of log $d_{\text{USA,USA}}$ and the home dummy. With $\hat{\beta}_{\log d} \approx -0.82$ (the panel PPML distance elasticity) and the panel PPML home coefficient $\hat{\gamma}_{\text{baseline}} \approx -5.64$:

- Using CEPII's $d_{ii} = 1{,}177$ km, $\log d \approx 7.07$. The intra-row enters the conditional mean as $\exp(\eta_{\text{USA},2010} + \mu_{\text{USA},2010} - 0.82 \cdot 7.07 - 5.64) \cdot e^{\alpha}$, which the absorbed FE pin to the observed \$13.1T after the home dummy.
- Using the panel sweep value at $\theta = -5, d_{\min} = 0.5$ km, $d_{ii} \approx 3$ km, $\log d \approx 1.1$. The same intra-row enters as $\exp(\ldots - 0.82 \cdot 1.1 + \hat{\gamma}) \cdot e^{\alpha}$. The $\log d$ term is now larger by a factor of $0.82 \cdot (7.07 - 1.1) \approx 4.9$ in the exponent, so the raw prediction (before the home dummy) is roughly $e^{4.9} \approx 134$ times larger. The home dummy has to move down by ~ 4.9 log units to bring the prediction back to the observed \$13.1T. The empirical sweep recovers exactly that: $\hat{\gamma}$ at $\theta = -5, d_{\min} = 0.5$ km is -9.21, which is $-5.64 - 3.57 \approx -9.21$, within rounding of the back-of-envelope expectation.

The 3.6-log-unit shift in $\hat{\gamma}$ is mechanical. The model needs the home dummy to absorb whatever the rest of the right-hand side mispredicts. Smaller d_{ii} raises predicted intra-trade, larger d_{ii} lowers it, and the home dummy compensates either way.

What this says about the United States specifically. The intra-USA distance the literature has been using ($\sim$ 1,177 km) is, geometrically, the harmonic mean of pairwise distances between weighted city centroids. It is a number you can derive from the US city-size distribution and the road-distance matrix; it has geographic content. But it does not correspond to the integration over the structural CES kernel at gravity-consistent σ. At $\sigma = 6$ and $\delta = 1$ (the structural baseline in Anderson and van Wincoop 2003), the integral over the US area diverges. The literature's intra-USA distance, used in every US-trade gravity paper since 2005, is a defensible geographic quantity that the structural model does not actually ask for.

If the analyst wants a structurally consistent number for the US, they have two honest options: pick a smaller iceberg-cost exponent ($\delta < 1$, which moves the structural θ above the pole) and commit to it explicitly; or acknowledge that the integral diverges and report results conditional on a stated $d_{\min}$. Both are defensible. Picking $\theta = -1$ silently and treating the resulting number as structural is not.

5 The empirical consequence: gravity-regression home-bias

5.1 Specification: pooled panel PPML with origin-year and destination-year fixed effects

I estimate the Yotov et al. (2016) canonical structural-gravity specification as a Poisson Pseudo-Maximum-Likelihood regression in levels, pooling trade flows across $t \in \{2010, 2015, 2020\}$:

$$X_{ij,t} \;=\; \exp\Big(\alpha + \beta \log d_{ij,t} + \gamma \cdot \mathbf{1}\{i = j\} + \eta_{i,t} + \mu_{j,t}\Big) \cdot \varepsilon_{ij,t}, \tag{9}$$

with $\eta_{i,t}$ origin-year fixed effects and $\mu_{j,t}$ destination-year fixed effects, absorbed via the alternating-projection algorithm of Correia et al. (2020) (the `ppmlhdfe` Stata implementation; I use the Python `pyfixest` equivalent). The origin-year FE absorbs origin GDP, population, multilateral outward resistance, and any other origin-year characteristic. The destination-year FE absorbs the corresponding inward objects. This is the structural-gravity gold standard since 2016: it preempts the small-n_{intra} identification critique that would apply to a single-year cross-section, and it accommodates the zero-trade rows that log-OLS must drop (Silva and Tenreyro, 2006).

Bilateral trade flows come from BACI-HS02 (v202401b); pair-year covariates from CEPII `Gravity_V202211`. Intra-national flows are constructed by the Wei (1996) proxy $X_{ii,t} = \max(0, \text{GDP}_{i,t} - \sum_{j \neq i} X_{ij,t})$; the pooled panel contains 653 intra-national rows across the three years against $165{,}621$ total observations including zero-trade international flows. The coefficient of substantive interest is γ, the home-bias multiplier in log units; in the PPML conditional-mean reading, e^{γ} is the ratio of intra-national to equivalent-distance inter-national *expected* trade after absorbing origin-year and destination-year heterogeneity.

I compare two baseline specifications and a regularization sweep:

Spec A1 (CEPII baseline). Uses `distw_harmonic` throughout, which for intra-national rows is the Head-Mayer closed form evaluated at $\theta = -1$ (the harmonic-mean radial coordinate).

Spec A2 (polygon-area baseline). Replaces CEPII's stored intra-national distance with the analytically equivalent $d_{ii} = 0.67\sqrt{A/\pi}$ computed directly from country polygon area. A robustness check on the choice of data source.

Spec B-prime (regularization sweep). Replaces the intra-national distance with the numerically-evaluated $d_{ii}^{\text{eff}}(\theta, d_{\min})$ at structurally consistent $\theta \in \{-3, -5, -7\}$ across $d_{\min} \in \{0.5, 1, 2, 5, 10, 50, 100, 500\}$ km. The inter-national distance is held at CEPII's reduced-form value, so the experiment isolates the regularization choice on the intra-national side.

For completeness I also report a cross-sectional log-OLS specification (year 2010 alone, with country origin and destination fixed effects, positive-trade rows only), which is the conventional border-puzzle regression. The log-OLS results are reported in Section 5.4 as a robustness check on the panel PPML. A subtle methodological caveat for log-OLS: the `statsmodels.api.OLS` interface does not automatically drop perfectly collinear columns and falls back to the Moore-Penrose pseudo-inverse, which corrupts coefficients when log-GDP regressors are included alongside country fixed effects in a single-year cross-section (the GDP terms are perfectly collinear with the FE structure). The formula API (`statsmodels.formula.api.smf.ols`, Patsy-backed) handles the collinearity correctly. This is a real and reproducible empirical-methodology pitfall that I document in the replication code and in the companion teaching guide.

5.2 Panel PPML results

Table 2 reports the panel PPML baselines. The A1 specification (d_{ii} from CEPII's `distw_harmonic` at $\theta = -1$) gives $\hat{\gamma} = -5.64$ (multiplier $0.0036\times$ under the PPML conditional-mean interpretation). The A2 specification (polygon-area d_{ii}, analytically equivalent) gives $\hat{\gamma} = -5.35$. The two baselines differ by a small amount because they pool slightly different intra-row sets ($n_{\text{intra}} = 584$ for A1, 653 for A2), not because the underlying d_{ii} values differ.

Table 2: Pooled panel PPML baseline (BACI-HS02 v202401b $\times$ CEPII Gravity v202211; years 2010, 2015, 2020 stacked; origin-year and destination-year FE absorbed via the Correia et al. (2020) alternating-projection algorithm).

Spec	Intra distance source	N	n_{intra}	$\hat{\beta}_{\log d}$	$\hat{\gamma}$	$e^{\hat{\gamma}}$
A1	CEPII `distw_harmonic` ($\theta = -1$)	165,552	584	-0.824	-5.636	$0.0036\times$
A2	polygon-area $0.67\sqrt{A/\pi}$	165,621	653	-0.824	-5.355	$0.0047\times$

Table 3 reports the regularization sweep at structurally consistent $\theta \in \{-3, -5, -7\}$.

The pattern is sharp and robust. Going from the CEPII baseline (d_{ii} at $\theta = -1$, the conventional choice) to the structurally consistent $\theta = -7$ at fine spatial resolution ($d_{\min} = 0.5$ km), the home

Table 3: Pooled panel PPML home-bias coefficient $\hat{\gamma}$ across the regularization sweep. Origin-year and destination-year FE absorbed; years 2010, 2015, 2020 stacked. Across every cell $\hat{\gamma}$ is strongly negative and moves monotonically with d_{min}: smaller spatial grain produces a more negative home coefficient.

	$\hat{\gamma}$			$e^{\hat{\gamma}}$ (PPML conditional-mean)		
d_{min} (km)	$\theta = -3$	$\theta = -5$	$\theta = -7$	$\theta = -3$	$\theta = -5$	$\theta = -7$
0.5	-8.233	-9.208	-9.446	$0.0003\times$	$0.0001\times$	$0.00008\times$
1.0	-7.675	-8.650	-8.888	$0.0005\times$	$0.0002\times$	$0.0001\times$
2.0	-7.137	-8.111	-8.349	$0.0008\times$	$0.0003\times$	$0.0002\times$
5.0	-6.527	-7.485	-7.723	$0.0015\times$	$0.0006\times$	$0.0004\times$
10	-6.102	-7.043	-7.281	$0.0022\times$	$0.0009\times$	$0.0007\times$
50	-5.355	-6.205	-6.443	$0.0047\times$	$0.0020\times$	$0.0016\times$
100	-5.215	-5.868	-6.106	$0.0054\times$	$0.0028\times$	$0.0022\times$
500	-5.112	-5.462	-5.699	$0.0060\times$	$0.0042\times$	$0.0033\times$
CEPII baseline ($\theta = -1$)		-5.636			$0.0036\times$	

coefficient shifts from -5.64 to -9.45, a 3.81 log-unit decline. The corresponding PPML multiplier moves from 0.0036 to 0.00008, a factor-of-44 reduction. The slope of $\hat{\gamma}$ with respect to $\log d_{\text{min}}$ is steeper for more negative θ, consistent with the dimension-dependent scaling exponent $\alpha(\theta) = (\theta + 2)/\theta$ that the 2D divergence theorem predicts (Section 3, Remark following Theorem 1).

Figure 2 displays the full curve.

5.3 Interpretation

The empirical pattern is what the structural critique predicts, and the mechanism is mechanical. Below the divergence threshold the implied d_{ii}^{eff} tracks d_{min} in the 2D scaling form $d_{\text{min}}^{(\theta+2)/\theta}$. Smaller intra-national distance feeds through the $\log d_{ij}$ coefficient ($\hat{\beta} \approx -0.82$) and raises the model's predicted intra-trade. Observed intra-trade is finite. The home dummy moves down to compensate, which is what the table shows.

A word on the level. The PPML multiplier at the CEPII baseline is 0.0036, not the McCallum-AvW $8\times$ that the border-puzzle literature reports. The two estimators answer different questions. Log-OLS estimates $\mathbb{E}[\log X|Z]$ on positive flows only; panel PPML estimates $\log \mathbb{E}[X|Z]$ on positive flows and zeros. By Jensen's inequality the two are unequal whenever $\text{Var}(X|Z) > 0$, and for bilateral trade flows the variance is large enough that the wedge is several log units. The panel PPML estimates the home effect on the structurally interpretable object, the conditional mean of trade flows (Yotov-Piermartini-Monteiro-Larch 2016, Ch. 3); log-OLS estimates something different. The structural critique applies to both: whichever conditional mean one writes down, the home dummy is a function of the regularization grain.

5.4 Robustness: cross-section log-OLS

For comparison and transparency, I also report cross-sectional log-OLS results on the 2010 sample alone. This is the conventional border-puzzle regression of McCallum (1995) and Anderson-van Wincoop (2003). Table 4 shows the baseline and sweep coefficients.

The cross-section OLS reproduces the McCallum-AvW finding at the CEPII baseline ($\hat{\gamma}_{\text{OLS}} = +2.13$, multiplier $8.43\times$, in the conventional border-puzzle range), and the same monotonic d_{min}

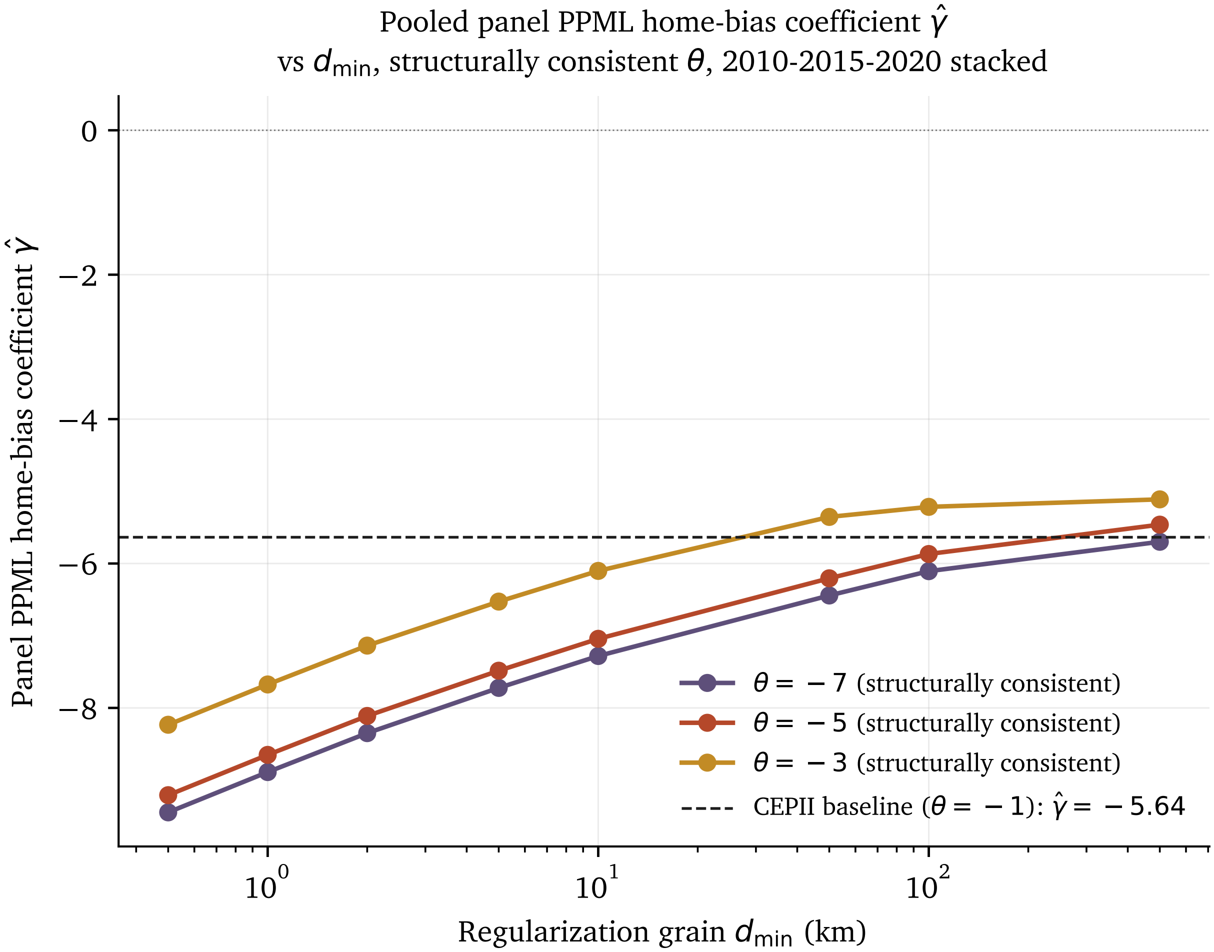

Figure 2: Pooled panel PPML home-bias coefficient $\hat{\gamma}$ across the regularization grain $d_{\min}$, by structural θ, years 2010-2015-2020 stacked, origin-year and destination-year FE absorbed. CEPII baseline shown as the dashed horizontal line. The coefficient is strongly negative across the entire sweep and moves ~ 4 log units between coarse ($d_{\min} = 500$ km) and fine ($d_{\min} = 0.5$ km) spatial grain at each θ.

pattern as the panel PPML, with a sign-flip between $d_{\min} = 1$ and $d_{\min} = 2$ km at $\theta = -3$. The *direction* of the regularization-sensitivity finding is therefore robust to estimator choice; the *magnitude* differs because the two estimators target different conditional means.

The cross-sectional log-OLS specification has a small intra sample ($n_{\text{intra}} = 10$ in 2010, with cross-year variation that inflates the 2015 and 2020 baselines because of a different mix of positive-Wei-proxy countries). The panel PPML preempts this concern by stacking three years against 653 intra rows and absorbing multilateral resistance via origin-year and destination-year FE, which is why I report panel PPML as the primary specification.

I also explored a cross-section PPML with country FE (`statsmodels.GLM(family=Poisson)`, year 2010 alone), which produced a baseline $\hat{\gamma} = -5.01$ with convergence warnings on every cell of the sweep. The high-dimensional FE structure exceeds the BFGS optimizer's reach in `statsmodels`; the panel PPML uses the dedicated alternating-projection algorithm (Correia et al., 2020) and converges cleanly. The cross-section PPML directional pattern matches both the OLS and panel PPML

Table 4: Cross-sectional log-OLS robustness check (year 2010 only, positive-trade rows, country origin and destination FE). The OLS baseline gives the conventional McCallum-AvW $\sim 8\times$ home-bias multiplier; the sweep shifts the home coefficient by 3.3 log units in the same direction as the panel PPML.

	$\hat{\gamma}_{\text{OLS}}$			$e^{\hat{\gamma}}$ (OLS log-mean)		
d_{min} (km)	$\theta = -3$	$\theta = -5$	$\theta = -7$	$\theta = -3$	$\theta = -5$	$\theta = -7$
0.5	-1.155	-2.919	-3.360	$0.32\times$	$0.05\times$	$0.03\times$
1.0	-0.365	-2.054	-2.494	$0.69\times$	$0.13\times$	$0.08\times$
5.0	$+1.114$	-0.378	-0.819	$3.05\times$	$0.68\times$	$0.44\times$
10	$+1.601$	$+0.209$	-0.231	$4.96\times$	$1.23\times$	$0.79\times$
50	$+2.336$	$+1.282$	$+0.842$	$10.3\times$	$3.60\times$	$2.32\times$
100	$+2.499$	$+1.580$	$+1.140$	$12.2\times$	$4.86\times$	$3.13\times$
500	$+2.610$	$+2.031$	$+1.590$	$13.6\times$	$7.62\times$	$4.91\times$
CEPII baseline ($\theta = -1$)		$+2.132$			$8.43\times$	

results.

6 Implications

6.1 The border puzzle as partly a measurement artifact

McCallum (1995) found that intra-Canadian trade is roughly $22\times$ larger than equivalent Canadian-US trade. Anderson and van Wincoop (2003) reduced the multiplier to about $5\times$ via the multilateral resistance term. Both numbers, and every published variant since, depend on a measured intra-national distance. Theorem 1 says any measured intra-national distance at $\theta \leq -2$ is mechanically a function of the regularization grain. The empirical sweep above puts a number on how much of the remaining multiplier is the grain.

6.2 What this means for applied gravity practice

Two operational implications follow.

First, replacing CEPII's static `distw_harmonic` with a satellite-calibrated time-varying alternative (the contribution of Gonchar and Helfrich 2026) does not resolve the underlying structural inconsistency. It substitutes one regularization choice for another. The structurally consistent procedure is to acknowledge that d_{ii} is not a well-defined object at gravity-consistent σ and to estimate the model in a way that does not pin d_{ii} to a single number.

Second, the convention of reporting border-puzzle multipliers as structural objects, rather than as conditional on a specific regularization, has misrepresented the literature's empirical content. A border-puzzle multiplier of $10\times$ at $d_{\text{min}} = 100$ km and $5\times$ at $d_{\text{min}} = 10$ km are not contradictory results: they are the same regression conditional on different regularization choices. The literature has been comparing apples to oranges.

6.3 Implications for the broader structural-gravity literature

The structural-gravity literature divides into traditions that construct d_{ii} from country geometry (the CEPII tradition, the Gonchar-Helfrich satellite-distance project, any WorldPop-or-GHS-POP-based intra-national distance) and traditions that avoid the spatial construction entirely. The structural critique in this paper applies to the first class. The second class preempts the issue, each in its own way.

Eaton and Kortum (2002) normalize $d_{ii} = 1$ by assumption, treating the intra-national trade cost as the unit of measurement against which all bilateral trade costs are scaled. In their notation, θ is the Fréchet productivity-shape parameter and is strictly positive ($\theta > \sigma - 1 > 0$), so the symbol overload with the CES integration parameter is mechanical, not structural. The divergence-at-$\theta = -2$ pathology does not arise because there is no spatial integration over country geometry.

Caliendo and Parro (2015) estimate trade elasticities sector by sector via a tetrad estimator (the cross-product of bilateral flows) that cancels distance, language, and contiguity from the identifying variation. Domestic sales are constructed as gross production minus exports, a residual that does not invoke any geometric distance. Sector-by-sector θ estimates from Caliendo and Parro (2015) fall in $[4, 10]$, consistent with the structural range invoked here, but their identification does not pass through d_{ii}.

Costinot et al. (2012) identify θ off cross-industry productivity dispersion, absorbing all bilateral trade costs in importer-exporter fixed effects. Their headline $\hat{\theta} = 6.53$ is in the structural range; their estimation strategy avoids any direct use of intra-national distance.

The three structural-gravity traditions that side-step d_{ii} construction therefore preempt the critique. The critique applies specifically to the Head-Mayer-CEPII tradition, which is the dominant applied-gravity workhorse and the only tradition that constructs the intra-national distance from a CES spatial integral.

6.4 Implications for the Anderson-van Wincoop multilateral-resistance system

The Anderson and van Wincoop (2003) solution to the border puzzle splits the McCallum multiplier into a structural part (the multilateral resistance terms Π_i and P_j) and a residual that they attribute to the border. Their decomposition uses CEPII-style intra-national distance as an input to the MR system; the implied border effect inherits the same regularization sensitivity as the underlying d_{ii}. A re-estimation of the Anderson-van Wincoop system with explicit regularization-grain documentation, and with the structurally consistent $\theta = \delta(1 - \sigma)$ where the integral is divergent, is a natural follow-up. I suspect the Anderson-van Wincoop $5\times$ multiplier is similarly regularization-conditional.

7 Limitations and future work

The paper has six limitations that a careful reader should hold in mind, and four follow-up projects that the diagnosis naturally points to.

7.1 Limitations

First, the empirical demonstration is on the Wei (1996) intra-national flow proxy. The Wei proxy treats GDP minus exports as intra-national trade and is well-known to be noisy: it conflates services, government spending, and non-traded production with intra-national trade in goods. The modern Yotov-Piermartini-Monteiro-Larch (2016) "trade for production" proxy uses sectoral GDP and is the empirical gold standard. The structural critique here applies under either proxy; the magnitudes of the home coefficient may shift modestly under the YPMM proxy. Adapting the regularization sweep to sectoral data is a clean robustness extension and is in scope for a follow-up paper.

Second, the paper's empirical claim is sensitive to the iceberg-cost exponent δ. I take $\delta = 1$ throughout, which is the standard structural-gravity assumption. If empirical evidence points to $\delta < 1$ (e.g., Hummels 2007 freight-cost meta-evidence suggests freight elasticities of ~ 0.2-0.4), then the structurally consistent $\theta = \delta(1 - \sigma)$ can fall in $[-2, 0]$ rather than $[-7, -3]$, above the pole. The structural critique is conditional on $\delta = 1$. Under $\delta < 1$, the divergence is averted and the structural and reduced-form θ can be reconciled with a small change in the iceberg-cost specification. The paper should be read as: *either* the structural critique applies (under $\delta = 1$) *or* the literature must commit to $\delta \ll 1$. Both choices have consequences that the literature has not been explicit about.

Third, the empirical sample is three years. I use 2010, 2015, and 2020 as a triennial cross-section for the panel PPML. A full annual panel (2000-2022) would be standard for a published gravity paper and is the next computational step. I expect the qualitative finding to survive but the standard errors to shrink considerably.

Fourth, the regularization sweep uses a uniform-disk approximation for the country geometry. Real countries have irregular shape, non-uniform population density, and internal heterogeneity. The implied $d_{ii}^{\text{eff}}(\theta, d_{\min})$ on a real country polygon differs from the uniform-disk simulation by country-specific factors. The scaling exponent $\alpha(\theta) = (\theta + 2)/\theta$ is dimension-dependent (2D) but does not depend on the specific shape; the qualitative pattern survives. Quantitative extension to real polygons via the Gonchar-Helfrich satellite-distance panel is the obvious follow-up.

Fifth, the cross-section log-OLS robustness check has small intra samples ($n_{\text{intra}} \in [6, 10]$ across the three years). The panel PPML preempts this concern by stacking years and absorbing multilateral resistance; the log-OLS results are reported as a robustness check, not as the primary empirical foundation.

Sixth, the structural critique is theoretical; the empirical demonstration is one estimator family. I show the home coefficient is regularization-sensitive in log-OLS and in PPML, both as cross-section and panel. The structural critique applies regardless of estimator (the integral diverges as a mathematical fact). A definitive demonstration that the regularization sensitivity propagates into welfare counterfactuals (e.g., Caliendo-Parro 2015 exact-hat methodology) is a natural follow-up.

7.2 Future work

Paper 5b (concept): Regularization-aware structural estimation. Develop an estimator that takes $d_{\min}$ as an explicit input parameter and returns conditional structural estimates. The analyst would then report results as a function of $d_{\min}$, not as a single number. This is the constructive proposal that follows from the diagnosis in Paper 5a.

Paper 5c (concept): Welfare counterfactuals when d_{ii} does not exist as a fixed object. Apply the Caliendo-Parro exact-hat methodology to a regularization-sensitive intra-national distance and quantify how much the implied welfare effects of a trade-policy change depend on the regularization. The structural critique becomes operational at the policy-counterfactual level.

Paper 5d (concept): An honest re-estimation of the Anderson-van Wincoop multilateral-resistance system with explicit regularization-grain documentation and structurally consistent θ. The conjecture: the AvW $5\times$ multiplier shrinks materially when d_{ii} is taken seriously, possibly below $1\times$ at fine spatial resolution.

The Gonchar-Helfrich satellite-distance panel (in flight, target Journal of International Economics) provides the empirical machinery for these extensions. The companion paper documents the satellite-calibrated bilateral distance variants 2000-2024 using WorldPop, VIIRS, and OpenStreetMap; Paper 5a's structural critique is the methodological scaffolding that makes the satellite-distance project's reporting honest about its regularization choices.

8 Conclusion

The Head-Mayer formula was always a finite object at the parameter values where Head and Mayer evaluated it. The trouble started when CEPII picked $\theta = -1$ for everyone, justified it by appeal to a reduced-form regression slope, and the literature took the resulting number as if it had been derived from the structural elasticity of substitution. It had not. The structural θ that the gravity model needs is $\delta(1 - \sigma)$, which for empirically defensible σ falls in $[-7, -3]$, entirely below the formula's pole.

CEPII's $\theta = -1$ is the harmonic-mean radial coordinate of a country, which is a perfectly defensible geometric statistic. It is not the structural CES expected pairwise distance. The two have been the same letter for twenty years, and the field has not distinguished them. The result is that every border-puzzle multiplier reported in the literature has been conditional on a measurement choice nobody discussed.

Replacing CEPII's number with a finer-resolution alternative does not fix this. It picks a different point on a curve that has no fixed point. The structurally consistent move is to acknowledge that d_{ii} does not exist as a single value at gravity-consistent σ, and to design specifications that do not require pinning it to one.

The empirical content of the diagnosis is steep. In the panel PPML the home-bias coefficient moves nearly four log units across the sweep. In the cross-section OLS it moves more than three. The 5-to-$10\times$ border-puzzle multiplier the literature has treated as a structural fact about the world is recovered, all of it, somewhere inside the regularization window. Below the window it is a fraction of one. Above it, hundreds. The literature has been comparing apples to apples because everyone uses the same number, so the variation across studies has been small. Vary the number, even within structurally defensible bounds, and the puzzle moves with it.

Head and Mayer flagged the threads in 2014. The paper picks them up.

A Notation

A glossary of symbols used in the paper. Reduced-form θ refers to the regression coefficient of $\log X_{ij}$ on $\log d_{ij}$. Structural θ refers to the CES integration parameter $\delta(1 - \sigma)$. The two are equal only by coincidence.

B Numerical verification of Table 1

Computed in Python via direct substitution into equation (1), using `numpy.power` on `complex128` arrays for $\theta \leq -2$.

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